





Building Automation

UNIT-1-INTRODUCTION TO BUILDING MANAGEMENT SYSTEM AND AUTOMATION

CO1:Understand Building Management system and Automation.

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WHAT IS BUILDING AUTOMATION?



A centralized, networked system of hardware and software that monitors and controls a building's facility systems (electricity, lighting, plumbing, HVAC, water supply, etc.).



When facilities are monitored and controlled in a seamless fashion, tenants enjoy a more comfortable working environment and facility management benefits from sustainable practices and reduced energy costs.

Introduction:

- A building management system (**BMS**), otherwise known as a building automation system (**BAS**), is a computer-based control system installed in buildings that controls and monitors the building's mechanical and electrical equipment such as ventilation, lighting, power systems, fire systems, and security.
- A BMS consists of software and hardware.
- The software program, usually configured in a hierarchical manner, can be proprietary, using such protocols as C-Bus , Profibus , and so on.
- Building management systems are most commonly implemented in large projects with extensive mechanical, HVAC, and electrical systems.



keep building climates within a specified range



monitor energy performance and device failures in all systems



light rooms according to an occupancy schedule



alarm facility managers in the event of a malfunction

THE FOUR CORE FUNCTIONS OF A BUILDING AUTOMATION SYSTEM



TO CONTROL THE BUILDING's ENVIRONMENT



TO MONITOR AND CORRECT SYSTEM PERFORMANCE



TO OPERATE SYSTEMS ACCORDING TO OCCUPANCY AND ENERGY DEMAND



TO ALERT OR SOUND ALARMS WHEN NEEDED

WHAT IS MEANT BY "CONTROLLED"?



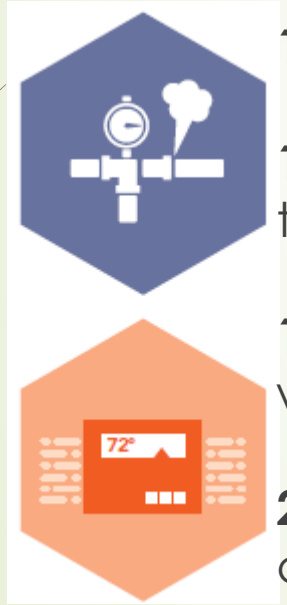
The key component in a BAS is the controller—a small, specialized computer with various applications

Controllers regulate the performance of various facilities within the building. Traditionally, this includes the following:

- Electrical systems
- Mechanical systems
- Heating, ventilation and air-conditioning systems
- Plumbing systems
- Lighting systems
- Surveillance Systems
- Security Systems

THE EVOLUTION OF SMART BUILDINGS

- Building automation has advanced just within our lifetime at a baffling rate.



1950: Automated buildings rely on pneumatic controls with compressed air.

1980s: Microprocessors move the industry from compressed air to analog then digital controls(nothing short of a revolution).

1990s: Open protocols allow controlled facilities to actually communicate with one another.

2000: Wireless technology allows components to communicate without cable attachments

TERMS TO UNDERSTAND



Building Management System and Building Control System: General terms for systems that control a building's facilities (not necessarily automation systems).



Direct Digital Control: Innovation in the '80s brought small, affordable microprocessors enabling a digital system's components to communicate



Application Programming Interface: The code that defines how two or more pieces of software communicate with one another.



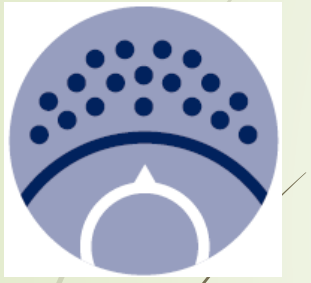
Building Automation System: A subset that can be a part of the larger BMS or BCS. Building management and building automation have so thoroughly overlapped that the terms are used interchangeably.



Energy Management System and Energy Management Control System: Energy consumption, metering, etc. There's enough overlap between what a BAS and an EMS do that we consider these terms synonymous.

HOW DO BUILDING AUTOMATION SYSTEMS WORK?

- **BASIC BAS HAS FIVE ESSENTIAL COMPONENTS:**



SENSORS
Devices that measure values such as CO2 output, temperature, humidity, daylight or room occupancy.



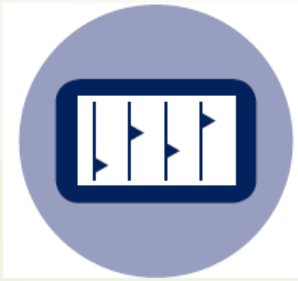
OUTPUT DEVICES
These carry out the commands from the controller. Examples: relays and actuators.



CONTROLLERS
The brains of the systems that take data from sensors and decide how the system will respond

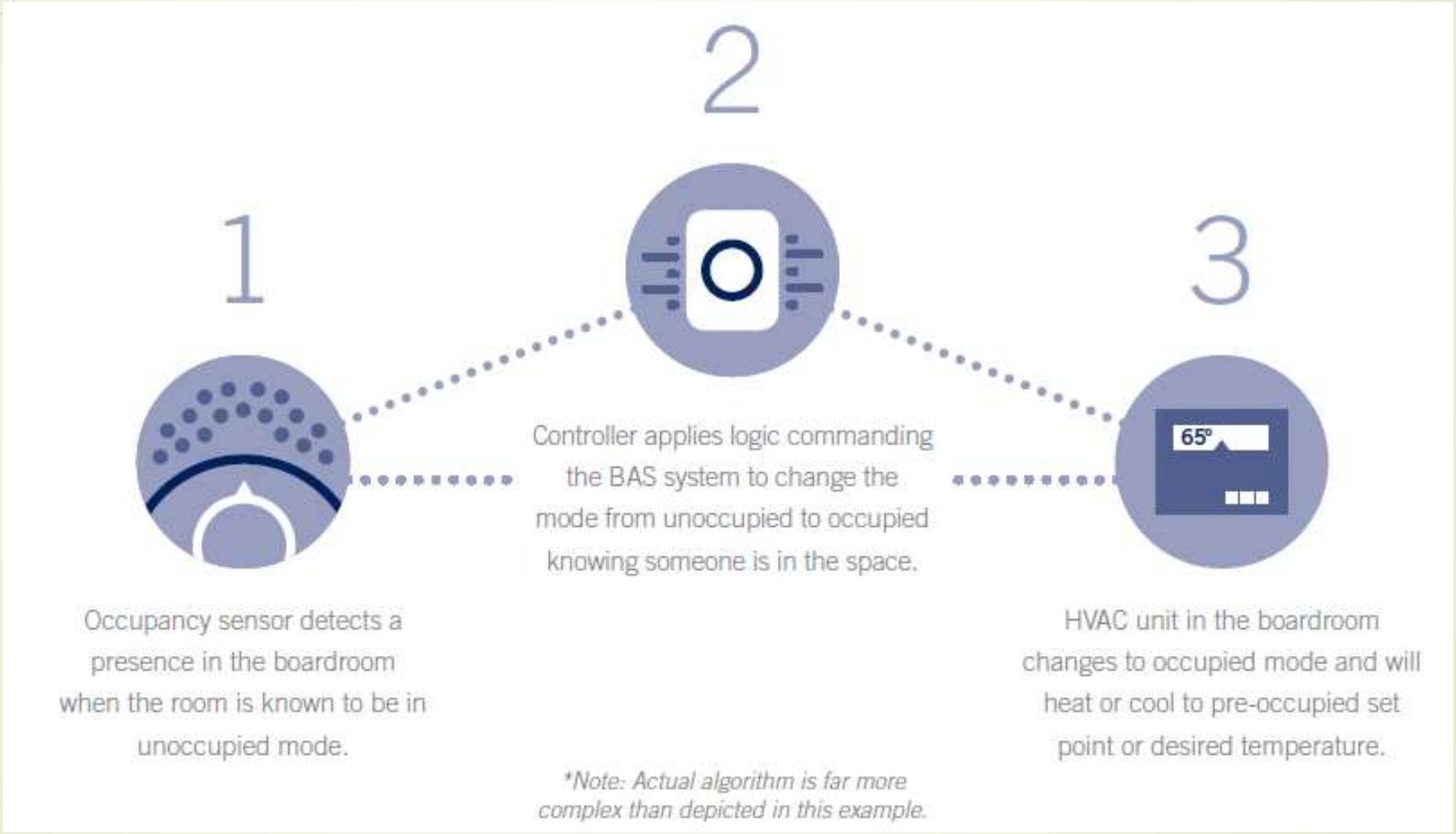


COMMUNICATIONS PROTOCOLS
The language spoken among the components of the BAS.
Example: BACnet.



DASHBOARD OR USER INTERFACE
Screens or interfaces humans use to interact with the BAS where building data are reported.

HOW A DDC LOOP WORKS:



WHY ARE BUILDING AUTOMATION SYSTEMS USEFUL?

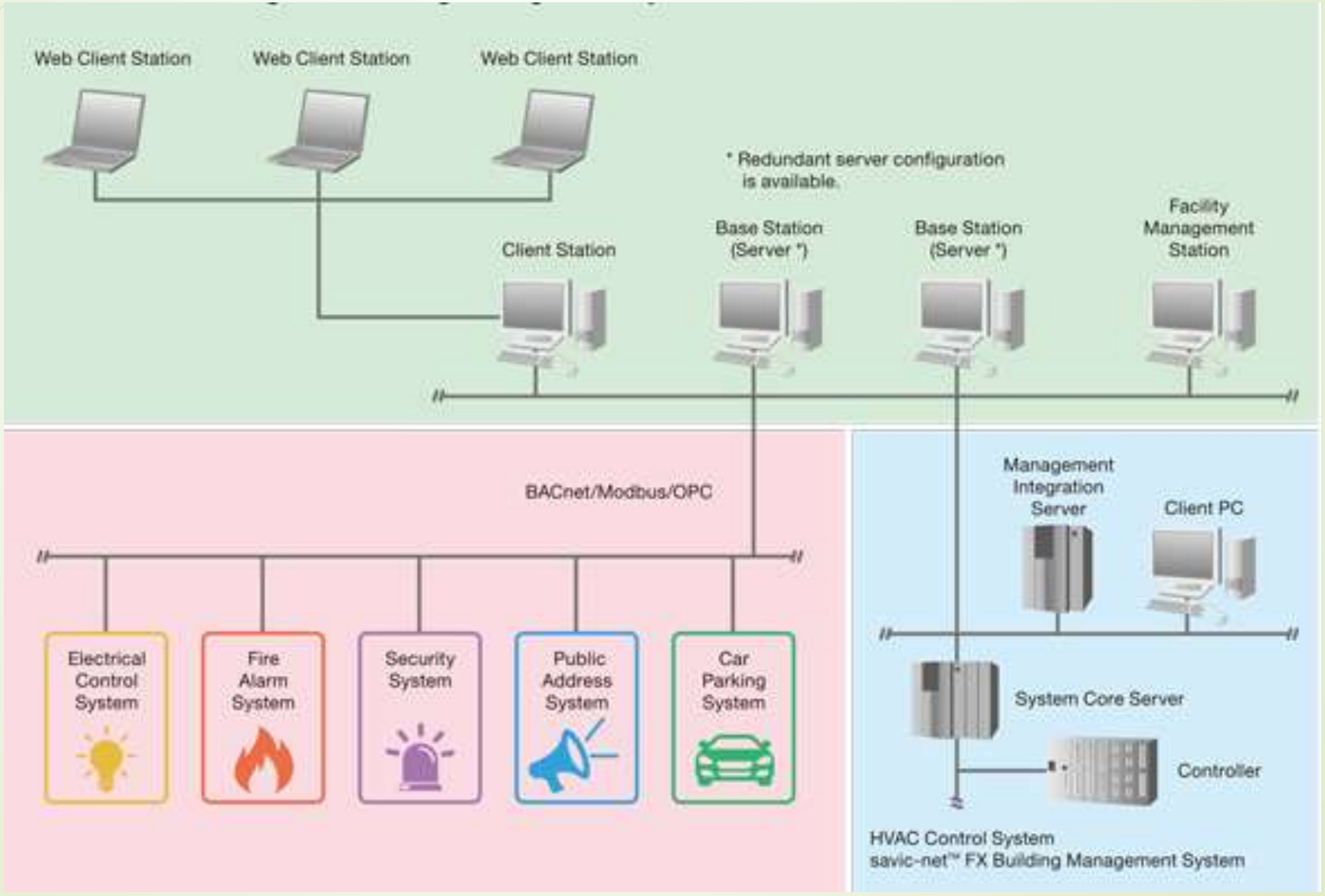


Smarter control of the building's environment keep occupants happier. It also reduces complaints and resolution time.

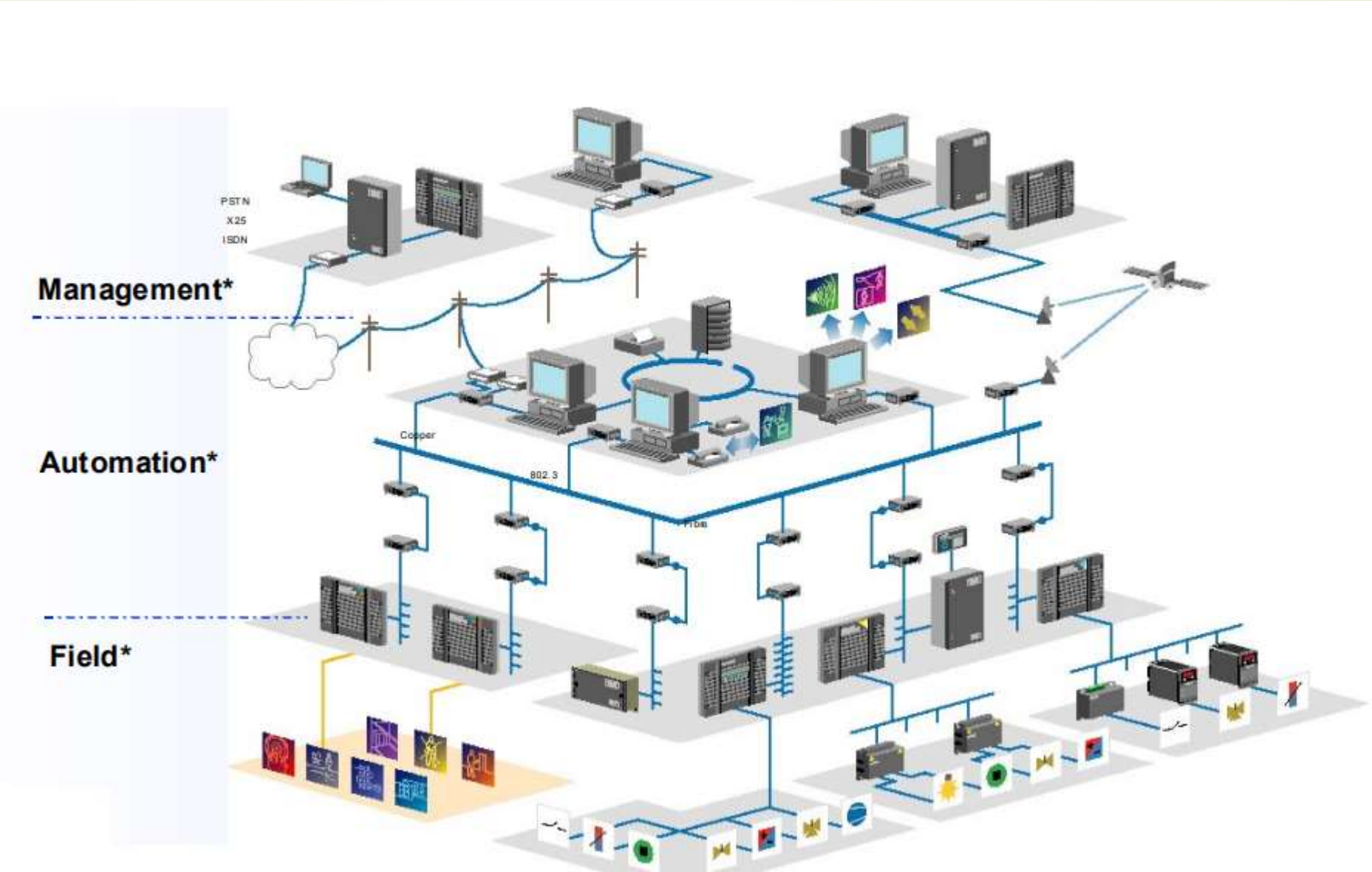
A BAS can save a building owner a significant amount of money on utility bills, since a more energy-efficient building simply costs less to run.

BAS efficiencies equate to a reduced environmental impact which means better air quality for everyone.

ARCHITECTURE OF BMS/BAS



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- **Upper level (Management Level).** Dispatching and administration as well as work with databases and statistical functions. At this level cooperation between personnel (operators, dispatchers etc) and system is performed, which is implemented by means of computer devices and SCADA system.
- **Middle level (Automation Level).** Automated management of functional processes. Basic components of this level are: main controllers, signal input-output units, and various switching equipment.
- **Low level (Field Level).** Level of terminals with input/output functions. This level includes sensors, actuating mechanisms, cabling between devices and low-middle levels.

Design of BMS

- **Define hardware point types:**

– Analogue - within a specified range

– Digital - On / Off or Open / Closed

– Metering - Water, Electricity via a Contact

- **Define Control Strategy:**

– Plant Control Strategy

– Software Interlocks

– Set Values

– Time Parameters

– Alarm Limits

Analogue Input

- **Analogue Input**

– 0-10Vdc

– 0-10 000 Ohms

– 0-20mA

- **Examples of A.I.**

– Temperature Sensor

– Humidity Sensor

– Pressure Transducer

– Light Level

– Solar Level

Analogue Output

- **Analogue Output**

– 0 to 10Vdc

- **Examples of A.O.**

– Electronic Actuators Valves / Dampers

– Speed Control

– Remote Setting Signal

Digital Input

- • **Digital Input** (Status or Binary Input)

– Clean Contact (Volt Free)

- • **Examples of D.I.**

– Air Flow Switch

– Door Contact

– Water Level Switch

Digital Output

- **Digital Output** (Binary Output)

– A Switch

– 240Vac Rated

– Override Switches

- **Examples of D.O.**

– Start / Stop Fan

– Start / Stop Chiller

– Switch On / Off Lights

Meter

- **Metered Input** (Pulse Input)

– A High Speed Pulse

– Up to 600 Pulses per Minute

- **Examples of Metering**

– Water Consumption

– Electricity Consumption

– Number of Persons Passing a Turnstile

Software Points

- **Used to define the Control Strategy**
- **Typical Software Point Types**

– TIM - Time Schedules

– SPA - Set Point

– CTL - PID Control Algorithms

– ROT - Rotation of Plant

– CAL - Calculation

– PRG - Programmable Freely Defined

– MD - Maximum Demand

Select Outstation

- Based on Hardware count
- • Match to appropriate Outstation
- • Fixed hardware or Modular
- • Allow spare capacity for future expansion

Components of BMS

- - **Centralized WorkStation Computer**

  With powerful user-friendly software.

  Used for everyday building operation.



- - **DDC Controllers**

  Micro-processor based

  Pre-configured / Freely programmable

  Controls the HVAC equipment of the building







- **Field devices**

  Temperature, Humidity, Pressure sensors

  Valves, Actuators







BMS Features

- **Man / Machine Interface**

- interact with the connected

-technical building equipment.

- user friendly

- for operators…

- engineers….

-and building managers

- **System Security**

- To prevent unauthorized use

- Password protection

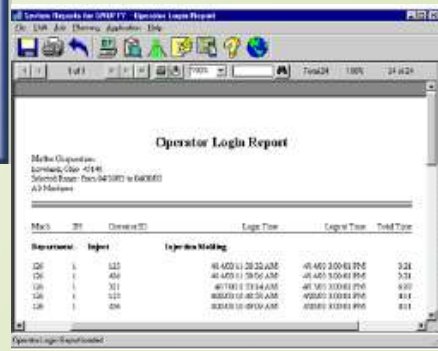
- Operator specific access

- Operator log summary









BMS Features



- **User-friendly data presentation**

-co-ordination of the flow of

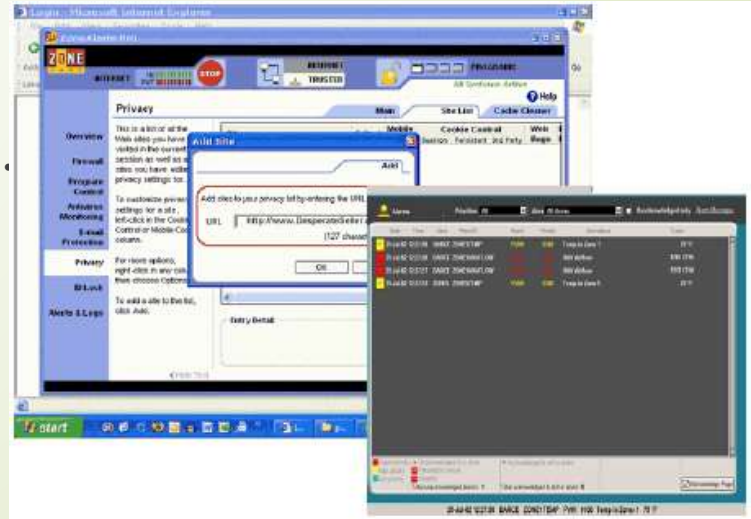
-information through the system by

-implementing customized graphics.

- floor plans of the building.

-graphical representation of the equipment.

- **Alarm Management**
- **the presentation in the sequence of importance and time of..**
  - potentially dangerous situations
  - process value deviations
- - **guiding the operator to take appropriate action through..**
  - audible and visual indications
  - Email, SMS
- - **alarm summary**
  - Time, date, priority and description



BMS Features

- **Reporting**

- present customized subsets of data

- actual or historical state

- export as a word or excel document or..

- to a printer..



- **Data Logging**

- The automatic gathering and

-storage of data from the field

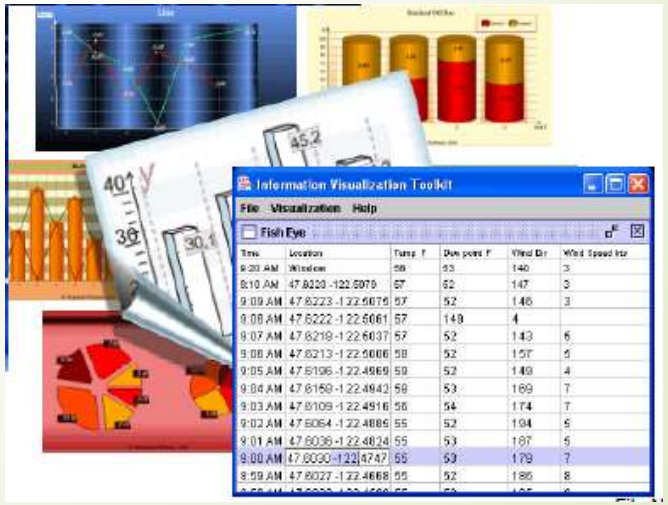
-equipment for later analysis and

-reporting

- Dynamically or historical

- Customized charts and graphs

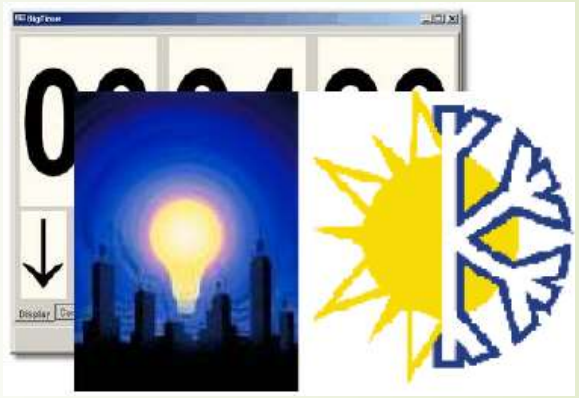
- Tabular reports

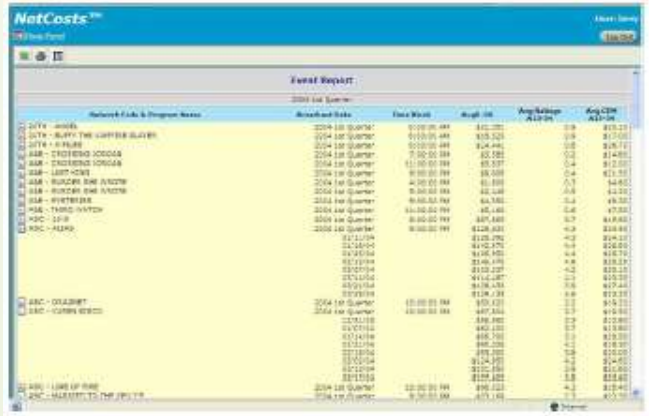


BMS Features

- **Time Scheduling**

- time-based start/stop of the equipment
- saves energy cost and efficient operation
- effective for lighting, occupancy control
- can be as daily, weekly, for holidays or events





- **Event recording**

- automatic logging of…
  - operator activities and commands
  - processes related to connected devices
  - workstations and printers



- **Remote connectivity**

- provide remote access to the system with full functionality through..
  - **local area network ,dial-up and internet via web browser**

BMS Benefits



- **Improved indoor environment quality**

- Comfortable living and working environment – domestic or commercial

- Better temperature and humidity control

- Good air quality



- **Faster response to ..**

- occupant needs

- end-user complaints

- trouble conditions

- **Maintenance Savings.**

- efficient control gives less wear and strain of mechanical equipment.

- provides longer life

- runtime monitoring alerts timely maintenance of equipment

- avoids expensive failures



BMS Benefits



- **Energy Savings**

- eliminates unnecessary system operation.

- accurate energy usage information

- helps you to take steps to reduce energy consumption like…

  - Optimum-Start
  - Night-Purging
  - Time-Scheduling

- **Consolidated facility control…**

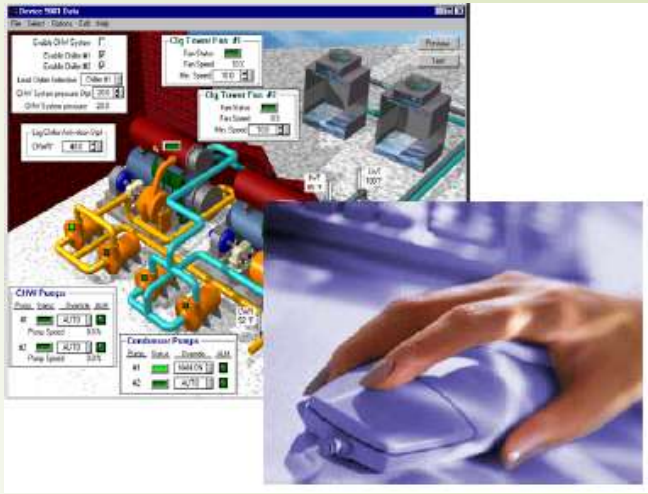
- One point centralized operation

- Simpler operation

- Reduces time and resources



BMS Benefits



- **Reduced operator training**

- on-screen instructions
- user-friendly graphic displays
- simpler operation programmed for routine and repetitive operation

- **Improved management reporting**



- Provides valuable real-time data
- Creates reports, charts…
- Critical information immediately sent to printers, emailed
- or sent via SMS

BMS Benefits

- **Timely and effective control**

- alerts the employees when your facility is not operating correctly

- reduce troubleshooting and down time.

- Remote access connectivity without site visits.



- **Performance Benchmarking**

- Facilitates the overall system performance measurement

- Comparison with set benchmarks

